Lodos's exact mode against Kamino: batched throughput

2026-11-03

scene	s	Δt (ms)	steps	Lodos's Newton solver	Kamino's iterations
cube_walled_5_polytope	3	2	1500	100 it, tol 1e-06, ls 50 it	20
cube_walled_5_sdf	3	2	1500	100 it, tol 1e-06, ls 50 it	–
arm_walled_5_polytope	3	2	1500	100 it, tol 1e-06, ls 50 it	20
arm_walled_5_sdf	3	2	1500	100 it, tol 1e-06, ls 50 it	–
shadow_grasp_polytope	5	2	2500	100 it, tol 1e-06, ls 50 it	200
shadow_grasp_sdf	5	2	2500	100 it, tol 1e-06, ls 50 it	–
shadow_walled_5_polytope	3	2	1500	100 it, tol 1e-06, ls 50 it	200
shadow_walled_5_sdf	3	2	1500	100 it, tol 1e-06, ls 50 it	–
nist_threading_m8	5	1.25	4000	100 it, tol 1e-06, ls 50 it	–
masonry_arch_polytope	2	4.17	480	100 it, tol 1e-06, ls 50 it	20
masonry_arch_sdf	2	4.17	480	100 it, tol 1e-06, ls 50 it	–
house_of_cards_polytope	2	4.17	480	100 it, tol 1e-06, ls 50 it	20
house_of_cards_sdf	2	4.17	480	100 it, tol 1e-06, ls 50 it	–
cube_walled_20_polytope	3	2	1500	100 it, tol 1e-06, ls 50 it	20
cube_walled_20_sdf	3	2	1500	100 it, tol 1e-06, ls 50 it	–
arm_walled_20_polytope	3	2	1500	100 it, tol 1e-06, ls 50 it	20
arm_walled_20_sdf	3	2	1500	100 it, tol 1e-06, ls 50 it	–
shadow_walled_20_polytope	3	2	1500	100 it, tol 1e-06, ls 50 it	left out
shadow_walled_20_sdf	3	2	1500	100 it, tol 1e-06, ls 50 it	–

Table 1: The scenes: the simulated seconds a run times, the step, the steps, Lodos's Newton solver (its iterations, tolerance and line-search iterations) and Kamino's PADMM iterations (–: a scene with Lodos surfaces, which Kamino does not run; left out: the twenty hands, where 16 worlds took 7.2 s a step).

scene	configuration	a world's capacities (1.5 times its peak over 16 starts)
cube_walled_5_polytope	Lodos, contact	nconmax 38, njmax 113, nccdmax 12
cube_walled_5_polytope	Kamino	contacts 54; its Delassus matrix 162 rows, 0.1 MB, so at most 81,827 worlds
cube_walled_5_sdf	Lodos, contact	nconmax 146, njmax 437; capacity_tile 0.116, capacity2 0.279
cube_walled_5_sdf	Lodos, cone	nconmax 20, njmax 59; capacity_tile 0.121, capacity2 0.269
cube_walled_5_sdf	Lodos, ray	nconmax 21, njmax 21; capacity_tile 0.105, capacity2 0.376
arm_walled_5_polytope	Lodos, contact	nconmax 225, njmax 140, nccdmax 21
arm_walled_5_polytope	Kamino	contacts 180; its Delassus matrix 540 rows, 1.2 MB, so at most 7,364 worlds
arm_walled_5_sdf	Lodos, contact	nconmax 27, njmax 81; capacity_tile 0.162, capacity2 0.0334
arm_walled_5_sdf	Lodos, cone	nconmax 20, njmax 59; capacity_tile 0.184, capacity2 0.0341
arm_walled_5_sdf	Lodos, ray	nconmax 26, njmax 26; capacity_tile 0.184, capacity2 0.044
shadow_grasp_polytope	Lodos, contact	nconmax 56, njmax 137, nccdmax 14
shadow_grasp_polytope	Kamino	contacts 21; its Delassus matrix 263 rows (200 of joints and limits), 0.3 MB, so at most 31,046 worlds

scene	configuration	a world's capacities (1.5 times its peak over 16 starts)
shadow_grasp_sdf	Lodos, contact	nconmax 15, njmax 83; capacity_tile 0.0354, capacity2 0.0755
shadow_grasp_sdf	Lodos, cone	nconmax 8, njmax 60; capacity_tile 0.0354, capacity2 0.0755
shadow_grasp_sdf	Lodos, ray	nconmax 8, njmax 45; capacity_tile 0.0548, capacity2 0.063
shadow_walled_5_polytope	Lodos, contact	nconmax 758, njmax 902, nccdmax 74
shadow_walled_5_polytope	Kamino	contacts 374; its Delassus matrix 2,002 rows (880 of joints and limits), 16 MB, so at most 535 worlds
shadow_walled_5_sdf	Lodos, contact	nconmax 330, njmax 1214; capacity_tile 0.0214, capacity2 0.0435
shadow_walled_5_sdf	Lodos, cone	nconmax 63, njmax 420; capacity_tile 0.0238, capacity2 0.0482
shadow_walled_5_sdf	Lodos, ray	nconmax 84, njmax 314; capacity_tile 0.0234, capacity2 0.0827
nist_threading_m8	Lodos, contact	nconmax 117, njmax 351, nccdmax 2; capacity_tile 0.259, capacity2 0.256
nist_threading_m8	Lodos, cone	nconmax 29, njmax 86, nccdmax 2; capacity_tile 0.258, capacity2 0.249
nist_threading_m8	Lodos, ray	nconmax 36, njmax 38, nccdmax 2; capacity_tile 0.448, capacity2 0.169
masonry_arch_polytope	Lodos, contact	nconmax 135, njmax 405, nccdmax 36
masonry_arch_polytope	Kamino	contacts 171; its Delassus matrix 525 rows (12 of joints and limits), 1.1 MB, so at most 7,791 worlds
masonry_arch_sdf	Lodos, contact	nconmax 1041, njmax 3123; capacity_tile 0.0443, capacity2 0.647
masonry_arch_sdf	Lodos, cone	nconmax 72, njmax 216; capacity_tile 0.0446, capacity2 0.646
masonry_arch_sdf	Lodos, ray	nconmax 72, njmax 72; capacity_tile 0.0452, capacity2 0.64
house_of_cards_polytope	Lodos, contact	nconmax 231, njmax 693, nccdmax 93
house_of_cards_polytope	Kamino	contacts 438; its Delassus matrix 1,314 rows, 6.9 MB, so at most 1,243 worlds
house_of_cards_sdf	Lodos, contact	nconmax 2123, njmax 6368; capacity_tile 0.0947, capacity2 0.0261
house_of_cards_sdf	Lodos, cone	nconmax 185, njmax 554; capacity_tile 0.132, capacity2 0.0275
house_of_cards_sdf	Lodos, ray	nconmax 371, njmax 371; capacity_tile 0.2, capacity2 0.0451
cube_walled_20_polytope	Lodos, contact	nconmax 119, njmax 356, nccdmax 30
cube_walled_20_polytope	Kamino	contacts 171; its Delassus matrix 513 rows, 1.1 MB, so at most 8,160 worlds
cube_walled_20_sdf	Lodos, contact	nconmax 939, njmax 2817; capacity_tile 0.0332, capacity2 0.298
cube_walled_20_sdf	Lodos, cone	nconmax 53, njmax 158; capacity_tile 0.032, capacity2 0.307
cube_walled_20_sdf	Lodos, ray	nconmax 53, njmax 53; capacity_tile 0.0347, capacity2 0.352
arm_walled_20_polytope	Lodos, contact	nconmax 756, njmax 369, nccdmax 62
arm_walled_20_polytope	Kamino	contacts 650; its Delassus matrix 1,950 rows, 15.2 MB, so at most 564 worlds
arm_walled_20_sdf	Lodos, contact	nconmax 96, njmax 288; capacity_tile 0.0563, capacity2 0.0315
arm_walled_20_sdf	Lodos, cone	nconmax 72, njmax 216; capacity_tile 0.0629, capacity2 0.0374
arm_walled_20_sdf	Lodos, ray	nconmax 78, njmax 78; capacity_tile 0.0595, capacity2 0.0323
shadow_walled_20_polytope	Lodos, contact	nconmax 2747, njmax 2580, nccdmax 179
shadow_walled_20_sdf	Lodos, contact	nconmax 549, njmax 2537; capacity_tile 0.00493, capacity2 0.0351
shadow_walled_20_sdf	Lodos, cone	nconmax 204, njmax 1577; capacity_tile 0.00474, capacity2 0.0376

scene	configuration	a world's capacities (1.5 times its peak over 16 starts)
shadow_walled_20_sdf	Lodos, ray	nconmax 252, njmax 1202; capacity_tile 0.00525, capacity2 0.0611

Table 2: Each configuration's capacities per world, times the worlds at a batch size: Lodos's MJWarp contacts, rows and CCD slots and its arenas as fractions of a world's worst case; Kamino's contacts, which size its dense Delassus matrix too, three rows a contact besides the joints' and limits'. One array holds every world's matrix, at most 2^{31} entries, Warp's limit, which caps Kamino's batch.

scene	Lodos, contact	Lodos, cone	Lodos, ray	Kamino
cube_walled_5_polytope	25 / 75	n/a	n/a	36
cube_walled_5_sdf	97 / 291	13 / 39	14 / 14	n/a
arm_walled_5_polytope	150 / 93	n/a	n/a	120
arm_walled_5_sdf	18 / 54	13 / 39	17 / 17	n/a
shadow_grasp_polytope	37 / 91	n/a	n/a	14
shadow_grasp_sdf	10 / 55	5 / 40	5 / 30	n/a
shadow_walled_5_polytope	505 / 601	n/a	n/a	249
shadow_walled_5_sdf	220 / 809	42 / 280	56 / 209	n/a
nist_threading_m8	78 / 234	19 / 57	24 / 25	n/a
masonry_arch_polytope	90 / 270	n/a	n/a	114
masonry_arch_sdf	694 / 2082	48 / 144	48 / 48	n/a
house_of_cards_polytope	154 / 462	n/a	n/a	292
house_of_cards_sdf	1415 / 4245	123 / 369	247 / 247	n/a
cube_walled_20_polytope	79 / 237	n/a	n/a	114
cube_walled_20_sdf	626 / 1878	35 / 105	35 / 35	n/a
arm_walled_20_polytope	504 / 246	n/a	n/a	433
arm_walled_20_sdf	64 / 192	48 / 144	52 / 52	n/a
shadow_walled_20_polytope	1831 / 1720	n/a	n/a	left out
shadow_walled_20_sdf	366 / 1691	136 / 1051	168 / 801	n/a

Table 3: A world's largest demand over the 16 starts: Lodos's contacts / constraint rows of the contact solve (MJWarp's contacts and rows, its surfaces' contacts among them, a group of them a contact); Kamino's contacts. n/a: a polytope scene has no Lodos surfaces, so no groups, and Kamino does not run the scenes with them.

scene	groups	B	default	no shared H.	fused Newton	kept
cube_walled_5_polytope	contact	1024	503k	502k	–	default
cube_walled_5_sdf	contact	1024	244k	244k	–	default
cube_walled_5_sdf	cone	1024	404k	404k	–	default
cube_walled_5_sdf	ray	1024	683k	682k	–	default
arm_walled_5_polytope	contact	1024	295k	294k	–	default
arm_walled_5_sdf	contact	1024	363k	364k	–	default
arm_walled_5_sdf	cone	1024	373k	374k	–	default
arm_walled_5_sdf	ray	1024	709k	708k	–	default
shadow_grasp_polytope	contact	1024	533k	568k	–	no shared H.
shadow_grasp_sdf	contact	1024	403k	420k	–	no shared H.
shadow_grasp_sdf	cone	1024	406k	440k	–	no shared H.
shadow_grasp_sdf	ray	1024	644k	677k	–	no shared H.
shadow_walled_5_polytope	contact	1024	124k	119k	–	default
shadow_walled_5_sdf	contact	1024	57.3k	55.8k	–	default
shadow_walled_5_sdf	cone	1024	52.5k	50k	–	default
shadow_walled_5_sdf	ray	1024	53.1k	49.1k	–	default

scene	groups	B	default	no shared H.	fused Newton	kept
nist_threading_m8	contact	1024	353k	352k	643k	fused Newton
nist_threading_m8	cone	1024	493k	489k	667k	fused Newton
nist_threading_m8	ray	1024	428k	423k	590k	fused Newton
masonry_arch_polytope	contact	1024	142k	126k	–	default
masonry_arch_sdf	contact	1024	44k	41.9k	–	default
masonry_arch_sdf	cone	1024	35.7k	32.9k	–	default
masonry_arch_sdf	ray	1024	19.9k	19.1k	–	default
house_of_cards_polytope	contact	1024	28.8k	29.3k	–	default
house_of_cards_sdf	contact	1024	8.11k	8.23k	–	default
house_of_cards_sdf	cone	1024	15.1k	15k	–	default
house_of_cards_sdf	ray	1024	13.1k	13.1k	–	default
cube_walled_20_polytope	contact	256	66.2k	60.4k	–	default
cube_walled_20_sdf	contact	256	35k	33.7k	–	default
cube_walled_20_sdf	cone	256	52.9k	47.4k	–	default
cube_walled_20_sdf	ray	256	77k	69.6k	–	default
arm_walled_20_polytope	contact	256	46.8k	43.8k	–	default
arm_walled_20_sdf	contact	256	59.4k	53.7k	–	default
arm_walled_20_sdf	cone	256	55.1k	49.7k	–	default
arm_walled_20_sdf	ray	256	79.7k	71k	–	default
shadow_walled_20_polytope	contact	256	3.66k	3.66k	–	default
shadow_walled_20_sdf	contact	256	1.67k	1.67k	–	default
shadow_walled_20_sdf	cone	256	1.31k	1.32k	–	default
shadow_walled_20_sdf	ray	256	1.19k	1.19k	–	default

Table 4: Lodos's execution options, world-steps/s at B worlds, from the capacity runs' half-way states: the default with MJWarp's Hessian summed in shared memory (on by default where it builds, on models of at most 144 degrees of freedom, padded), without it (no shared H.), and the fused Newton solve; an option is kept where it is 3 % faster ("–": refused, the fused solve takes models of at most 16 degrees of freedom).

scene	B	default	sparse J.	RCM	both	kept
cube_walled_5_polytope	1024	421k	420k	338k	323k	default
arm_walled_5_polytope	1024	75.3k	70.5k	65.3k	57.8k	default
shadow_grasp_polytope	1024	11.1k	11.8k	11.1k	11.9k	both
shadow_walled_5_polytope	256	250	487	OOM	OOM	sparse J.
masonry_arch_polytope	1024	13.2k	24k	14.7k	28.6k	both
house_of_cards_polytope	512	5.29k	8.88k	5.19k	9.14k	both
cube_walled_20_polytope	256	71.2k	70.6k	55.2k	55k	default
arm_walled_20_polytope	256	9.29k	7.85k	OOM	OOM	default

Table 5: Kamino's execution options, world-steps/s at B worlds, from the capacity runs' half-way states: the default (dense Jacobians, its Delassus matrices factored by blocked Cholesky), sparse Jacobians (sparse J.), the factorization after a reverse Cuthill–McKee reordering (RCM), and both: the same solve each. An option is kept where it is 3 % faster. B is halved from 1024 until the default fits. OOM: out of memory.

scene	cap	best	iterations	converged	ladder
cube_walled_5_polytope	20	589k @4096	20	0.366 %	dropped at 8192
arm_walled_5_polytope	20	58.3k @128	20	0 %	dropped at 256
shadow_grasp_polytope	200	11.1k @512	200	0 %	dropped at 1024
shadow_walled_5_polytope	200	501 @256	200	0 %	out of memory at 512
masonry_arch_polytope	20	35.3k @2048	20	0 %	out of memory at 4096

scene	cap	best	iterations	converged	ladder
house_of_cards_polytope	20	9.34k @512	20	0 %	out of memory at 1024
cube_walled_20_polytope	20	76.6k @4096	20	0 %	past 2^{31} entries at 8192
arm_walled_20_polytope	20	9.66k @256	20	0 %	out of memory at 512

Table 6: Kamino's solves: its PADMM iterations' cap, its best throughput (world-steps/s at the batch size after @), and in that run's last step its worlds' mean PADMM iterations and the share of them whose solve met its tolerances; how its ladder ended.

scene	Lodos, contact	Lodos, cone	Lodos, ray	Kamino	ratio
cube_walled_5_polytope	711k @8192	n/a	n/a	589k @4096	1.21
cube_walled_5_sdf	225k @2048	596k @8192	1.08M @4096	n/a	1.83‡
arm_walled_5_polytope	408k @8192	n/a	n/a	58.3k @128	6.99
arm_walled_5_sdf	523k @4096	544k @4096	1.12M @8192	n/a	19.26‡
shadow_grasp_polytope	983k @8192	n/a	n/a	11.1k @512	88.33
shadow_grasp_sdf	660k @8192	675k @8192	1.02M @8192†	n/a	60.64‡
shadow_walled_5_polytope	103k @8192	n/a	n/a	501 @256	205.50
shadow_walled_5_sdf	65.3k @4096	69.4k @8192	74.4k @4096	n/a	148.50‡
nist_threading_m8	1.12M @8192	1.25M @8192	1.45M @8192	n/a	–
masonry_arch_polytope	175k @16384	n/a	n/a	35.3k @2048	4.97
masonry_arch_sdf	44.3k @2048	34.8k @1024	21.4k @1024	n/a	1.26‡
house_of_cards_polytope	34.2k @8192	n/a	n/a	9.34k @512	3.66
house_of_cards_sdf	7.89k @2048	18.6k @4096	16.8k @2048	n/a	1.99‡
cube_walled_20_polytope	189k @16384	n/a	n/a	76.6k @4096	2.47
cube_walled_20_sdf	61k @4096	148k @16384	199k @8192	n/a	2.60‡
arm_walled_20_polytope	102k @16384	n/a	n/a	9.66k @256	10.59
arm_walled_20_sdf	143k @16384	142k @16384	216k @16384	n/a	22.31‡
shadow_walled_20_polytope	2.62k @1024	n/a	n/a	left out	–
shadow_walled_20_sdf	1.74k @512	1.58k @512	1.56k @512	n/a	–

Table 7: The best throughput of each configuration, world-steps/s at the batch size after @, and the ratio of Lodos's to Kamino's on the same bodies, Lodos's the fastest of its configurations that lost no world. ‡: against Kamino on the scene's polytope version, which collides the bodies as convex hulls, the scene having Lodos surfaces. †: the run lost worlds, non-finite or escaped (Table 9). n/a: a polytope scene has no Lodos surfaces, so no groups, and Kamino does not run the scenes with them.

scene	configuration	world-steps/s	B	× real time	× Kamino
cube_walled_5_polytope	Lodos, contact	711k	8192	1 420	1.21
	Kamino	589k	4096	1 180	1.00
cube_walled_5_sdf	Lodos, contact	225k	2048	449	0.38‡
	Lodos, cone	596k	8192	1 190	1.01‡
	Lodos, ray	1.08M	4096	2 150	1.83‡
arm_walled_5_polytope	Lodos, contact	408k	8192	815	6.99
	Kamino	58.3k	128	117	1.00
arm_walled_5_sdf	Lodos, contact	523k	4096	1 050	8.98‡
	Lodos, cone	544k	4096	1 090	9.33‡
	Lodos, ray	1.12M	8192	2 250	19.26‡
shadow_grasp_polytope	Lodos, contact	983k	8192	1 970	88.33
	Kamino	11.1k	512	22.3	1.00

scene	configuration	world-steps/s	B	× real time	× Kamino
`shadow_grasp_sdf`	Lodos, contact	660k	8192	1 320	59.33‡
	Lodos, cone	675k	8192	1 350	60.64‡
	Lodos, ray	1.02M†	8192	2 050	92.00‡
`shadow_walled_5_polytope`	Lodos, contact	103k	8192	206	205.50
	Kamino	501	256	1	1.00
`shadow_walled_5_sdf`	Lodos, contact	65.3k	4096	131	130.38‡
	Lodos, cone	69.4k	8192	139	138.63‡
	Lodos, ray	74.4k	4096	149	148.50‡
`nist_threading_m8`	Lodos, contact	1.12M	8192	1 390	–
	Lodos, cone	1.25M	8192	1 570	–
	Lodos, ray	1.45M	8192	1 810	–
`masonry_arch_polytope`	Lodos, contact	175k	16384	730	4.97
	Kamino	35.3k	2048	147	1.00
`masonry_arch_sdf`	Lodos, contact	44.3k	2048	185	1.26‡
	Lodos, cone	34.8k	1024	145	0.99‡
	Lodos, ray	21.4k	1024	89.1	0.61‡
`house_of_cards_polytope`	Lodos, contact	34.2k	8192	142	3.66
	Kamino	9.34k	512	38.9	1.00
`house_of_cards_sdf`	Lodos, contact	7.89k	2048	32.9	0.84‡
	Lodos, cone	18.6k	4096	77.3	1.99‡
	Lodos, ray	16.8k	2048	70.1	1.80‡
`cube_walled_20_polytope`	Lodos, contact	189k	16384	378	2.47
	Kamino	76.6k	4096	153	1.00
`cube_walled_20_sdf`	Lodos, contact	61k	4096	122	0.80‡
	Lodos, cone	148k	16384	295	1.93‡
	Lodos, ray	199k	8192	399	2.60‡
`arm_walled_20_polytope`	Lodos, contact	102k	16384	205	10.59
	Kamino	9.66k	256	19.3	1.00
`arm_walled_20_sdf`	Lodos, contact	143k	16384	286	14.78‡
	Lodos, cone	142k	16384	283	14.67‡
	Lodos, ray	216k	16384	431	22.31‡
`shadow_walled_20_polytope`	Lodos, contact	2.62k	1024	5.24	–
`shadow_walled_20_sdf`	Lodos, contact	1.74k	512	3.48	–
	Lodos, cone	1.58k	512	3.16	–
	Lodos, ray	1.56k	512	3.11	–

Table 8: Each configuration's best throughput: world-steps/s at a batch of B worlds; how many times real time it ran, the seconds simulated per second summed over the worlds (world-steps/s times the step: 2 ms, the threading's 1.25 ms, the arch's and the house of cards' 4.17 ms; each world's own is that over B); and its ratio to Kamino's best on the same bodies. ‡: against Kamino on the scene's polytope version, as convex hulls. †: the run lost worlds, non-finite or escaped (Table 9).

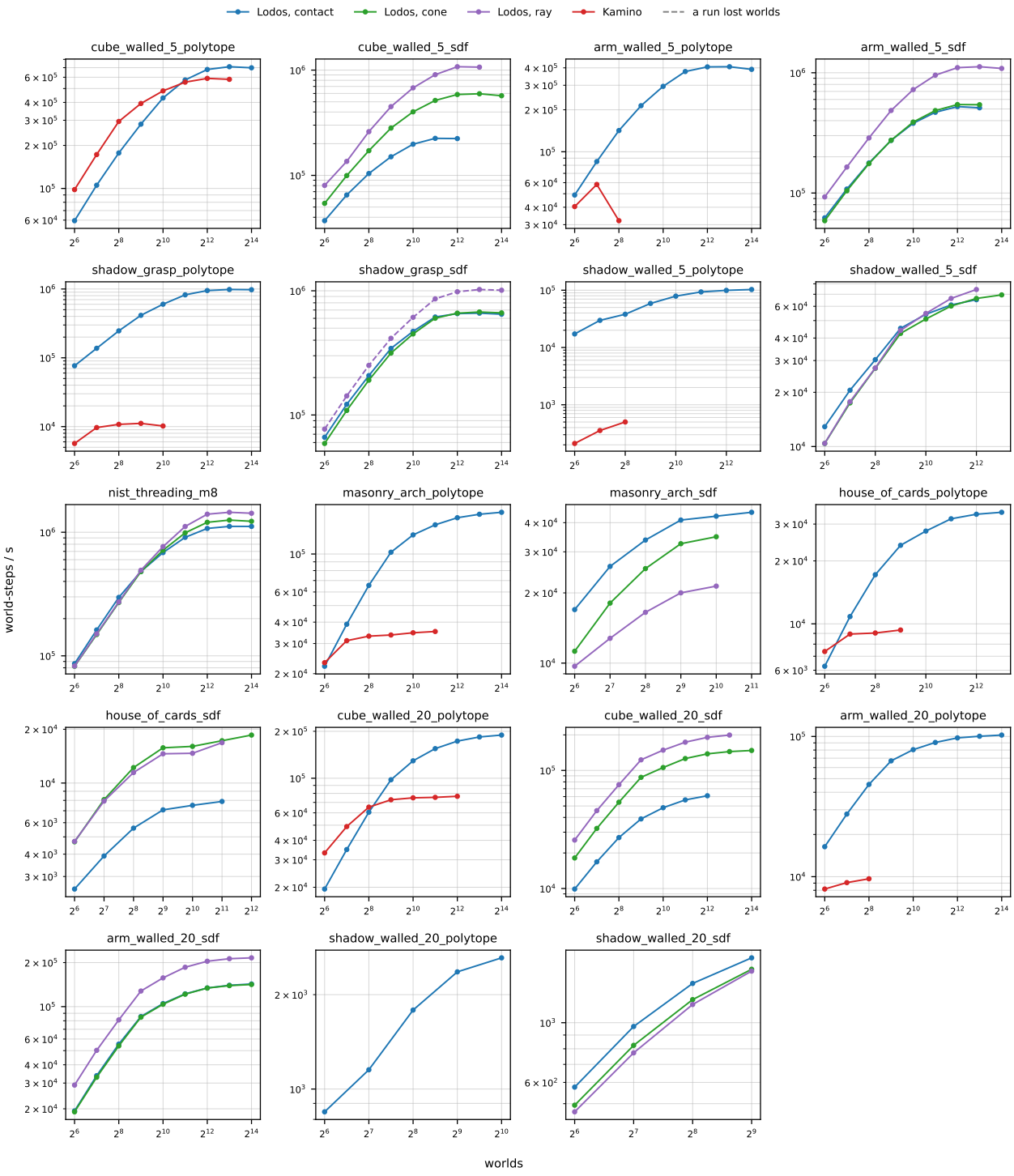


Figure 1: World-steps per second against the batch size, each configuration's ladder; dashed where a run lost worlds.

scene	configuration	world-steps/s by batch size; how the ladder ended; worlds lost at the best
cube_walled_5_polytope	Lodos, contact	64: 59.5k, 128: 105k, 256: 177k, 512: 282k, 1024: 430k, 2048: 574k, 4096: 680k, 8192: 711k, 16384: 699k; dropped at 16384; 0 non-finite, 0 escaped
cube_walled_5_polytope	Kamino	64: 98.1k, 128: 173k, 256: 294k, 512: 393k, 1024: 482k, 2048: 554k, 4096: 589k, 8192: 580k; dropped at 8192; 0 non-finite, 0 escaped
cube_walled_5_sdf	Lodos, contact	64: 37.1k, 128: 65.1k, 256: 104k, 512: 150k, 1024: 198k, 2048: 225k, 4096: 224k; dropped at 4096; 0 non-finite, 0 escaped; 1 redone after an overflow
cube_walled_5_sdf	Lodos, cone	64: 54.2k, 128: 99.5k, 256: 172k, 512: 281k, 1024: 402k, 2048: 515k, 4096: 587k, 8192: 596k, 16384: 571k; dropped at 16384; 0 non-finite, 0 escaped
cube_walled_5_sdf	Lodos, ray	64: 80.3k, 128: 136k, 256: 260k, 512: 450k, 1024: 678k, 2048: 905k, 4096: 1.08M, 8192: 1.07M; dropped at 8192; 0 non-finite, 0 escaped
arm_walled_5_polytope	Lodos, contact	64: 48.9k, 128: 85k, 256: 142k, 512: 214k, 1024: 295k, 2048: 376k, 4096: 406k, 8192: 408k, 16384: 390k; dropped at 16384; 0 non-finite, 0 escaped; 1 redone after an overflow
arm_walled_5_polytope	Kamino	64: 40.4k, 128: 58.3k, 256: 32k; dropped at 256; 0 non-finite, 0 escaped; 1 redone after an overflow
arm_walled_5_sdf	Lodos, contact	64: 62.2k, 128: 108k, 256: 178k, 512: 274k, 1024: 380k, 2048: 469k, 4096: 523k, 8192: 511k; dropped at 8192; 0 non-finite, 0 escaped
arm_walled_5_sdf	Lodos, cone	64: 58.9k, 128: 104k, 256: 176k, 512: 274k, 1024: 389k, 2048: 485k, 4096: 544k, 8192: 542k; dropped at 8192; 0 non-finite, 0 escaped
arm_walled_5_sdf	Lodos, ray	64: 92.8k, 128: 164k, 256: 287k, 512: 486k, 1024: 724k, 2048: 955k, 4096: 1.1M, 8192: 1.12M, 16384: 1.09M; dropped at 16384; 0 non-finite, 0 escaped
shadow_grasp_polytope	Lodos, contact	64: 76.6k, 128: 137k, 256: 246k, 512: 414k, 1024: 601k, 2048: 822k, 4096: 949k, 8192: 983k, 16384: 976k; dropped at 16384; 0 non-finite, 0 escaped
shadow_grasp_polytope	Kamino	64: 5.68k, 128: 9.68k, 256: 10.8k, 512: 11.1k, 1024: 10.2k; dropped at 1024; 0 non-finite, 0 escaped
shadow_grasp_sdf	Lodos, contact	64: 66.1k, 128: 122k, 256: 207k, 512: 344k, 1024: 471k, 2048: 615k, 4096: 655k, 8192: 660k, 16384: 648k; dropped at 16384; 0 non-finite, 0 escaped
shadow_grasp_sdf	Lodos, cone	64: 59k, 128: 109k, 256: 191k, 512: 315k, 1024: 449k, 2048: 601k, 4096: 660k, 8192: 675k, 16384: 664k; dropped at 16384; 0 non-finite, 0 escaped
shadow_grasp_sdf	Lodos, ray	64: 77k, 128: 142k, 256: 251k, 512: 414k, 1024: 611k, 2048: 861k, 4096: 982k, 8192: 1.02M, 16384: 1.01M; dropped at 16384; 0 non-finite, 8192 escaped
shadow_walled_5_polytope	Lodos, contact	64: 17.2k, 128: 29.8k, 256: 38.1k, 512: 58.8k, 1024: 78.8k, 2048: 93.4k, 4096: 99.4k, 8192: 103k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped; 1 redone after an overflow
shadow_walled_5_polytope	Kamino	64: 211, 128: 355, 256: 501, 512: –; out of memory at 512; 0 non-finite, 0 escaped
shadow_walled_5_sdf	Lodos, contact	64: 12.9k, 128: 20.5k, 256: 30.3k, 512: 45.3k, 1024: 54.2k, 2048: 61.1k, 4096: 65.3k, 8192: –; out of memory at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow

scene	configuration	world-steps/s by batch size; how the ladder ended; worlds lost at the best
shadow_walled_5_sdf	Lodos, cone	64: 10.4k, 128: 17.5k, 256: 27.2k, 512: 42.4k, 1024: 51k, 2048: 60.2k, 4096: 66.3k, 8192: 69.4k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped
shadow_walled_5_sdf	Lodos, ray	64: 10.4k, 128: 17.7k, 256: 27.3k, 512: 44.1k, 1024: 54.8k, 2048: 66.3k, 4096: 74.4k, 8192: –; out of memory at 8192; 0 non-finite, 0 escaped
nist_threading_m8	Lodos, contact	64: 86.4k, 128: 162k, 256: 297k, 512: 481k, 1024: 685k, 2048: 909k, 4096: 1.07M, 8192: 1.12M, 16384: 1.11M; dropped at 16384; 0 non-finite, 0 escaped
nist_threading_m8	Lodos, cone	64: 82.4k, 128: 149k, 256: 270k, 512: 480k, 1024: 715k, 2048: 989k, 4096: 1.2M, 8192: 1.25M, 16384: 1.23M; dropped at 16384; 0 non-finite, 0 escaped
nist_threading_m8	Lodos, ray	64: 82.7k, 128: 151k, 256: 273k, 512: 492k, 1024: 765k, 2048: 1.11M, 4096: 1.4M, 8192: 1.45M, 16384: 1.42M; dropped at 16384; 0 non-finite, 0 escaped
masonry_arch_polytope	Lodos, contact	64: 22.1k, 128: 38.9k, 256: 65.6k, 512: 102k, 1024: 129k, 2048: 148k, 4096: 163k, 8192: 171k, 16384: 175k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
masonry_arch_polytope	Kamino	64: 23.2k, 128: 31.2k, 256: 33.1k, 512: 33.7k, 1024: 34.7k, 2048: 35.3k, 4096: –; out of memory at 4096; 0 non-finite, 0 escaped
masonry_arch_sdf	Lodos, contact	64: 17k, 128: 26k, 256: 33.7k, 512: 41k, 1024: 42.6k, 2048: 44.3k, 4096: –; out of memory at 4096; 0 non-finite, 0 escaped
masonry_arch_sdf	Lodos, cone	64: 11.2k, 128: 18.1k, 256: 25.4k, 512: 32.5k, 1024: 34.8k, 2048: –; out of memory at 2048; 0 non-finite, 0 escaped
masonry_arch_sdf	Lodos, ray	64: 9.69k, 128: 12.8k, 256: 16.5k, 512: 20k, 1024: 21.4k, 2048: –; out of memory at 2048; 0 non-finite, 0 escaped
house_of_cards_polytope	Lodos, contact	64: 6.27k, 128: 10.8k, 256: 17.2k, 512: 23.7k, 1024: 27.8k, 2048: 31.8k, 4096: 33.4k, 8192: 34.2k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped
house_of_cards_polytope	Kamino	64: 7.37k, 128: 8.93k, 256: 9.04k, 512: 9.34k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped
house_of_cards_sdf	Lodos, contact	64: 2.55k, 128: 3.91k, 256: 5.59k, 512: 7.08k, 1024: 7.5k, 2048: 7.89k, 4096: –; out of memory at 4096; 0 non-finite, 0 escaped
house_of_cards_sdf	Lodos, cone	64: 4.69k, 128: 8.08k, 256: 12.2k, 512: 15.8k, 1024: 16k, 2048: 17.3k, 4096: 18.6k, 8192: –; out of memory at 8192; 0 non-finite, 0 escaped
house_of_cards_sdf	Lodos, ray	64: 4.72k, 128: 7.93k, 256: 11.5k, 512: 14.6k, 1024: 14.7k, 2048: 16.8k, 4096: –; out of memory at 4096; 0 non-finite, 0 escaped
cube_walled_20_polytope	Lodos, contact	64: 19.5k, 128: 34.9k, 256: 60.6k, 512: 97.8k, 1024: 129k, 2048: 155k, 4096: 173k, 8192: 184k, 16384: 189k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
cube_walled_20_polytope	Kamino	64: 33.2k, 128: 49k, 256: 65.3k, 512: 72.8k, 1024: 74.9k, 2048: 75.4k, 4096: 76.6k, 8192: –; past 2^{31} entries at 8192; 0 non-finite, 0 escaped
cube_walled_20_sdf	Lodos, contact	64: 9.9k, 128: 16.8k, 256: 27k, 512: 38.8k, 1024: 48.3k, 2048: 56.3k, 4096: 61k, 8192: –; out of memory at 8192; 0 non-finite, 0 escaped; 1 redone after an overflow
cube_walled_20_sdf	Lodos, cone	64: 18.1k, 128: 32.3k, 256: 53.8k, 512: 87.4k, 1024: 106k, 2048: 126k, 4096: 138k, 8192: 144k, 16384: 148k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped

scene	configuration	world-steps/s by batch size; how the ladder ended; worlds lost at the best
cube_walled_20_sdf	Lodos, ray	64: 25.7k, 128: 45.6k, 256: 75.6k, 512: 123k, 1024: 149k, 2048: 173k, 4096: 191k, 8192: 199k, 16384: –; out of memory at 16384; 0 non-finite, 0 escaped
arm_walled_20_polytope	Lodos, contact	64: 16.4k, 128: 28k, 256: 45.5k, 512: 67k, 1024: 80.5k, 2048: 90.6k, 4096: 97.7k, 8192: 100k, 16384: 102k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
arm_walled_20_polytope	Kamino	64: 8.16k, 128: 9.07k, 256: 9.66k, 512: –; out of memory at 512; 0 non-finite, 0 escaped
arm_walled_20_sdf	Lodos, contact	64: 19.3k, 128: 33.7k, 256: 55.4k, 512: 85.5k, 1024: 105k, 2048: 122k, 4096: 134k, 8192: 140k, 16384: 143k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
arm_walled_20_sdf	Lodos, cone	64: 19.1k, 128: 32.9k, 256: 53.9k, 512: 84.7k, 1024: 104k, 2048: 122k, 4096: 134k, 8192: 139k, 16384: 142k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
arm_walled_20_sdf	Lodos, ray	64: 29.1k, 128: 50.2k, 256: 81.1k, 512: 128k, 1024: 157k, 2048: 186k, 4096: 204k, 8192: 213k, 16384: 216k, 32768: –; out of memory at 32768; 0 non-finite, 0 escaped
shadow_walled_20_polytope	Lodos, contact	64: 846, 128: 1.15k, 256: 1.79k, 512: 2.36k, 1024: 2.62k, 2048: –; out of memory at 2048; 0 non-finite, 0 escaped
shadow_walled_20_sdf	Lodos, contact	64: 576, 128: 968, 256: 1.4k, 512: 1.74k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped
shadow_walled_20_sdf	Lodos, cone	64: 494, 128: 824, 256: 1.22k, 512: 1.58k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped
shadow_walled_20_sdf	Lodos, ray	64: 466, 128: 773, 256: 1.17k, 512: 1.56k, 1024: –; out of memory at 1024; 0 non-finite, 0 escaped

Table 9: Every ladder: each batch size's world-steps/s, from 64 doubled, how it ended, and the worlds the best run lost.